Quiz- 1
CHE109
Summer 2024
Time-20min

Name:
ID:
SEC:
Date of Exam: 20 11 24
Marks: 10

Answer all the problems:

1. (a) State the law of multiple proportions. (1+2)
   (b) Two compounds of sulfur and oxygen are analyzed. In Compound P, 50 g of sulfur reacts with 50 g of oxygen. In Compound Q, 50 g of sulfur reacts with 100 g of oxygen. Demonstrate how this data satisfies the law of multiple proportions.
2. An electron in a hydrogen atom transitions to the n=2 state, emitting a photon with a wavelength of 486 nm. Calculate the initial quantum number for this transition. ($h=6.626\times10^{-34}$ J, $c=3.0\times10^{8}$ m/s, $R_H=2.18\times10^{-18}$ J) (3)
3. A tennis ball of mass 0.058 kg moves at 40 m/s. Does it exhibit a de Broglie wavelength? Explain why this wavelength is not observed in practical situations. (1)

   What is the de Broglie wavelength of an alpha particle ($m=6.64\times10^{-27}$) moving at a speed of 2.0×10^{6} m/s? Compare this wavelength with that of an electron ($m=9.11\times10^{-31}$) moving at the same speed. (3)

Please start answering after the following line